STRUCTURED SLOT MEMORY FOR MASKED-EVIDENCE WORLD-MODEL INTERFACES

Anonymous authors
Paper under double-blind review

ABSTRACT

Finite-context world models can observe a decisive visual cue and later act as if it was never seen. We treat this as an interface problem rather than an architecture label: old evidence must survive beyond the legal context window and must also be exposed through a frozen host representation in a form a later consumer can use. A valid memory claim must therefore separate four questions: whether the cue lies outside the legal window, whether it is retained after the window expires, whether the frozen interface exposes it, and whether a bounded controller can act on it. We instantiate this ladder with a self-supervised masked-evidence JEPA objective and a compact slot-memory writer. Training never sees cue labels, cue crops, or key-frame annotations; salient patch sets are mined from the rendered stream, withheld from the visible context, and predicted from a causal memory state, while cue labels enter only afterwards for a fail-closed audit under matched full, reset, and no-state conditions. Across a suite of OGBench render decks, controlled evidence ages, and seeds, the slot interface clears every registered retention gate while matched controls stay at chance, and the largest margins over recurrent GRU, LSTM, and Mamba-lite writers appear exactly where evidence is old or spread across object-like structure. Trained on a mixture of delays, the same interface generalizes to unseen delays and scales to far longer horizons at constant per-step memory and compute, since the writer is an explicitly bounded streaming state that evicts old tokens. The advantage survives matching the carrier capacity and adding structured state-space, gated-slot, and retrieval memories, does not depend on a colour shortcut when the cue identity is decorrelated from colour, and transfers into a frozen LeWorldModel host, where the injected memory exposes old evidence in the host's own predictive output beyond its context window. The contribution is a scoped audit-and-method result: persistent old-evidence memory in latent world models is a property of the target–carrier–host–consumer interface, not a universal property of any single recurrent or state-space module.

1 INTRODUCTION

Finite-context world models often fail in a way that is easy to miss: they can observe a decisive cue early and later produce a representation that behaves as if the cue was never present. This matters for physical intelligence because action-relevant evidence is rarely synchronized with the moment of action. A robot can read a sign before turning away, see a door state before entering a hallway, or observe an object identity before that object becomes occluded. If the interface to a frozen host exposes only a recent window, then locally plausible prediction is not enough; the old evidence must survive long enough, in a usable form, to change a later decision.

We study this failure as an *interface* problem rather than a raw memory capacity problem. The question is not only whether a hidden state can memorize something. The stricter question is whether old visual evidence, after it has left the legal context window, is (i) *retained* in a causal memory state, (ii) *exposed* through a host-facing representation, and (iii) *usable* by a bounded downstream controller. These three claims are routinely conflated. A model may store old evidence internally yet fail to expose it in its output. A linear readout may decode a label through a shortcut if the current endpoint still leaks the cue. A native controller may succeed only because it received the true goal rather than the memory-selected one. Our evaluation is designed to close each of these loopholes with matched controls.

We introduce a *masked-evidence slot-memory interface* (Figure 1). Training is self-supervised: it never receives cue labels, cue crops, or manually selected key frames. A target miner divides the causal render stream into temporal bins, scores visual saliency, selects salient patch sets, and removes them from the visible stream. A compact slot transformer must then predict the stop-gradient latent targets of those hidden patches from a small persistent memory state. Cue labels enter only *after* training, when a post-hoc ridge readout tests whether the learned memory exposes the old cue under matched full, reset, and no-state conditions.

The central empirical point is structural rather than merely numerical. Under the same caches, target mining, patch target encoder, loss, optimizer, and readout, replacing the slot writer with GRU, LSTM, or Mamba-lite recurrence weakens long-delay exposure. The recurrent baselines often solve short, waypoint-like rows, which is useful evidence that the task is not impossible; their failure appears when the cue is old or distributed across object-like visual evidence. Slots help because the target itself is a *set* of visual facts, and a memory set is a better carrier for that target than a single recurrent vector.

Concretely, on twelve OGBench render decks, three controlled evidence ages (4, 8, 15), and three seeds, the slot writer passes every registered retention cell (36/36 environment-age rows, 108/108 seed cells) with mean full-memory balanced accuracy 0.983 while reset and no-state controls stay at 0.250 and 0.248. The strongest recurrent baseline reaches 0.803 mean full accuracy and $22/36$ rows; at age 15 the slot writer scores 0.962 against 0.669, 0.350, and 0.683. Where audited fixed controllers exist, memory-selected targets execute at 0.957 mean success across $27/36$ rows.

Contributions.

- A **fail-closed audit** for old-evidence memory in frozen world-model interfaces, separating retention, host exposure, and bounded use with matched reset / no-state controls.
- A **label-free masked patch-set JEPA objective** that trains a slot memory writer from automatically mined salient patches, with full pseudocode (Algorithms 1–2).
- A **controlled architecture comparison** across twelve OGBench decks, three evidence ages, and three seeds, isolating the writer format while holding every other component fixed, extended with capacity-matched and structured-memory baselines (SlotSSM-style, gated slot attention, top-k retrieval, cached attention).
- A **bounded streaming formulation** $M_t = f_\theta(M_{t-1}, Z_{t-K+1:t})$ with token eviction and constant per-step memory/compute, plus a **delay-generalization** and **age-scaling** study to age 128.
- A **frozen-LeWM integration** showing the same interface exposes old evidence in the host's own output beyond its context window, and a **shortcut-broken benchmark** confirming retention of object identity rather than colour.

Our contribution is best read as a *fail-closed evaluation protocol plus a controlled demonstration that target–carrier structural alignment improves host-visible old-evidence exposure*, rather than a wholly new slot-memory architecture: slot memory (Jiang et al., 2024; Zhang et al., 2024; Anonymous, 2026), masked JEPA prediction (Anonymous, 2026), and memory-augmented world models (Anonymous, 2025; LeWorldModel Team, 2026) each exist; the novelty is the interface audit that separates storage, host exposure, and use and shows where structural alignment matters.

2 Related Work

World models under memory pressure. Latent world models compress observations into predictive state for control (Hafner et al., 2023). Recall-to-Imagine makes long-term memory and credit assignment explicit bottlenecks for model-based RL (Samsami et al., 2024). Diffusion and video world models face a related pressure: short observation windows can produce plausible futures that drift from earlier evidence. LoopNav turns this into looped spatial-consistency tests (Lian et al., 2025); StateSpaceDiffuser, long-context state-space video models, and memory-expert diffusion world models add state-space or expert memory paths to preserve temporal consistency across long rollouts (Savov et al., 2025; Po et al., 2025; Stapf et al., 2026). We ask a narrower but stricter interface question: not whether a generator looks temporally consistent, but whether old evidence that is illegal for the current context is still exposed in a frozen host-facing representation.

Partial observability and memory in RL. Partially observable RL has long treated memory as a substitute for unavailable Markov state. PO-Dreamer adds memory guidance to world models under partial observability (Li et al., 2025), and memory traces replace fixed observation windows with compact, analyzable history summaries (Eberhard et al., 2025). These works optimize agents or value estimates. In contrast, our audit isolates a representation-interface claim: the downstream readout receives the same endpoint under full, reset, and no-state conditions, so success must come from retained old evidence rather than a policy learning to exploit a different observation stream.

Sequence memory and learned state. GRU and LSTM recurrence remain standard baselines for compressing sequence history (Cho et al., 2014; Hochreiter and Schmidhuber, 1997). Mamba and Mamba-2 scale selective state-space recurrence and connect it to attention-like computation (Gu and Dao, 2023; Dao and Gu, 2024). Infini-attention, test-time-training layers, and Titans push the fixed-context problem from compressive attention, self-supervised hidden-state updates, and neural long-term memory (Munkhdalai et al., 2024; Sun et al., 2024; Behrouz et al., 2025). SlotSSM and Gated Slot Attention are closest to our writer because they make sequence memory modular or slot-like rather than a single monolithic vector (Jiang et al., 2024; Zhang et al., 2024). These works support the broad argument that fixed-window attention is insufficient for long-horizon evidence; they do not, by themselves, answer whether a memory state exposes the task-relevant old cue under leakage controls, which is exactly what we hold fixed and test.

Structured and embodied memory. Object-centric world models show that decision-relevant visual details are often small, structured, and easily washed out by reconstruction losses (Zhang et al., 2025). SOLD and language-guided object-centric world models show that slot latents can serve as relational state interfaces for manipulation (Mosbach et al., 2025; Jeong et al., 2025). Embodied-memory systems such as KARMA, Retrieval-Augmented Embodied Agents, Memo, and 3DLLM-Mem separate short-term working context from longer experience or spatial-temporal memory (Wang et al., 2024; Zhu et al., 2024; Gupta et al., 2025; Hu et al., 2025); agent-memory work such as Memory-R1 and ReMemR1 studies what to store, update, and retrieve (Yan et al., 2026; Shi et al., 2026). Evaluation work such as RULER and LongMemEval warns that easy retrieval tests overstate real memory ability (Hsieh et al., 2024; Wu et al., 2025). Our slot memory is aligned with this structured direction, but the supervision and claim differ: we train from label-free masked visual patch sets and reveal labels only to audit whether stored evidence is linearly accessible and physically useful.

3 Problem Formulation

Let $x_{0:T}$ denote rendered frames and $a_{0:T-1}$ actions. A cue $c \in \{0, 1, 2, 3\}$ is visible at an early time and determines a later target label $y = c$. At readout time t, the frozen host sees only a recent context window of length K; the cue lies strictly outside that window. We call the delay between the cue's last visible frame and the readout boundary the *evidence age* α. Rather than committing to a few fixed delays, we treat α as a continuous variable and sample it along a grid; the values $\{4, 8, 15\}$ are simply convenient reference points on that axis, and the delay study (Section 6) sweeps α up to 128. We report retention as a function of α rather than as pass/fail at a hand-picked age.

Audit conditions. The audit has three conditions that share the same held-out episodes, endpoint, and readout, differing only in the memory state fed to the readout. The *full* condition uses the causal memory state M_t produced from the full legal stream. The *reset* condition observes the same endpoint and recent context but clears persistent memory at the context boundary. The *no-state* condition removes persistent memory entirely. Because all three share the endpoint, any information recoverable from current pixels, the action trace, class imbalance, or a forgiving readout raises *all* conditions together and therefore fails the audit.

Claim ladder. We report a four-rung ladder in which each rung is interpreted only after the previous one passes. **(1) Demand:** the decisive cue must sit outside the legal context window (ages $4/8/15$). **(2) Retention:** full memory must expose the old cue on held-out episodes. **(3) Controls:** reset and no-state features must stay near chance. **(4) Use:** executed success is reported only where a fixed local controller exists; missing rows are marked controller-unavailable rather than counted as successes.

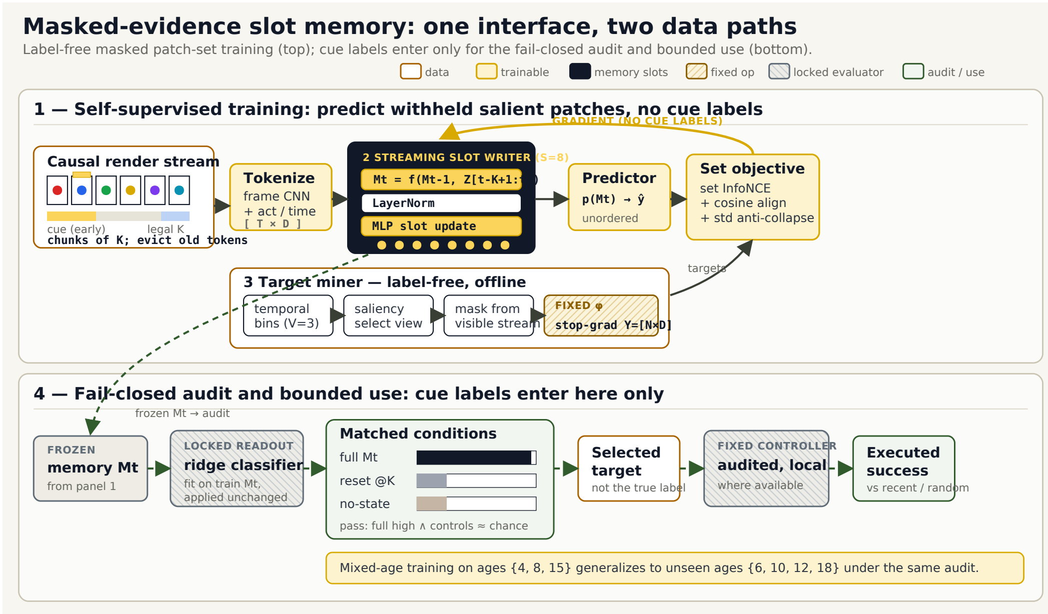

Figure 1: Masked-evidence slot-memory interface and its two data paths. A causal render stream is tokenized and written into eight memory slots M_t (stages 1–3). A label-free target miner withholds salient patch sets and encodes them with a fixed encoder ϕ (stage 4); the slot writer predicts those hidden targets through a set-prediction loss (stage 5, yellow gradient path). Cue labels are never used in training; a post-hoc ridge readout audits M_t under matched full / reset / no-state conditions (stage 6, dashed eval-only path).

Formally, a seed cell passes when

$$\mathrm{BAcc}_{\text{full}} \geq 0.750 \quad \wedge \quad \mathrm{BAcc}_{\text{reset}} \leq 0.350 \quad \wedge \quad \mathrm{BAcc}_{\text{no-state}} \leq 0.350, \tag{1}$$

and an environment-age row passes only when all three seeds pass. Chance balanced accuracy is 0.25 for the four-way cue.

4 Method

Figure 1 is the compact method contract. A trajectory cache fixes frames, actions, time indices, and train/validation splits for every method so that only the memory writer changes. We describe each component and then give training and audit pseudocode.

4.1 Inputs and tokenization

For each episode the model receives the legal stream: rendered frames $x_{0:t}$, actions $a_{0:t-1}$, and integer time indices. A frame CNN maps each frame to a token $z_i \in \mathbb{R}^D$ with $D = 160$; learned action and time embeddings are added, giving a token sequence $Z = [z_0, \ldots, z_t] \in \mathbb{R}^{t \times D}$. The cue frames are inside the mined-and-hidden set at training time and outside the length-K window at audit time, so the model never has a trivially legal copy of the answer.

4.2 Masked patch-set target

The training signal is old, visual, and label-free. For each episode the target miner splits the causal history $[0, t]$ into $V{=}3$ temporal bins. Within each bin it scores patch saliency (local color contrast and novelty against a running mean), samples one salient view per bin proportionally to saliency, and extracts a set of high-saliency patches. The selected patches are *removed from the visible stream* and encoded by a fixed target encoder ϕ into pooled color/texture latents,

$$Y = \{y_j\}_{j=1}^{N}, \qquad y_j = \mathrm{stopgrad}\big(\phi(\mathrm{patch}_j)\big) \in \mathbb{R}^D, \tag{2}$$

where ϕ applies a fixed $4{\times}4$ adaptive pool, mean-centering, and a saturation channel followed by a frozen random projection. Because ϕ is fixed and generic, the miner cannot smuggle a semantic label into the target: it only sees color and texture structure. Crucially, the target is a *set*, which keeps object-level, spatially distributed structure intact rather than collapsing it into a single frame reconstruction.

4.3 Slot writer

The writer maintains $S{=}8$ normalized memory slots. Slots act as queries in a compact multi-head cross-attention over the token sequence, followed by LayerNorm and an MLP update; stacking these blocks yields the causal memory state $M_t = [m_1, \dots, m_S] \in \mathbb{R}^{S\times D}$. Writing one block is

$$\tilde{M} = M + \mathrm{MHA}\big(Q{=}W_q M,\ K{=}W_k Z,\ V{=}W_v Z\big), \tag{3}$$

$$M' = \tilde{M} + \mathrm{MLP}\big(\mathrm{LN}(\tilde{M})\big), \tag{4}$$

with $H{=}4$ heads.

4.4 Bounded streaming memory

A causal transformer over the full prefix is causal but not bounded: it can re-read every historical token at readout. To make the memory a genuine bounded persistent state we use a chunked recurrence. The legal stream is ingested in consecutive chunks of at most K tokens, and the slots attend only over the previous slot state concatenated with the current chunk; the chunk tokens are then *evicted*:

$$M_t \ = \ f_\theta\big(M_{t-1},\ Z_{t-K+1:t}\big), \qquad \text{keys/values} = [\, M_{t-1}\, ;\, Z_{t-K+1:t}\,], \tag{5}$$

so all tokens before $t-K+1$ are physically deleted and inaccessible. Peak activation memory and per-step compute are therefore constant in sequence length (Section 6.1); with K larger than the sequence the update reduces exactly to the one-shot writer in Eq. (4), which we keep as an ablation. Because both the memory and the target are sets, the model is not forced to squeeze every old fact into one recurrent vector or one hand-indexed slot; different slots can specialize to different mined views.

4.5 Set-prediction objective

A prediction head p maps slots to predicted target latents $\hat{Y} = p(M_t)$. Predictions are matched to the mined target set Y with a set loss that combines contrastive, cosine, and anti-collapse terms:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{InfoNCE}}(\hat{Y}, Y)}_{\text{retain the right facts}} + \lambda_{\cos}\ \underbrace{\big(1 - \overline{\cos}(\hat{Y}, Y)\big)}_{\text{align latents}} + \lambda_{\text{std}}\ \underbrace{\text{std-reg}(\hat{Y})}_{\text{anti-collapse}}. \tag{6}$$

The InfoNCE term uses in-batch negatives so the memory must keep the correct old view separable from other episodes' views; the cosine term aligns matched latents; the standard-deviation regularizer prevents slot collapse. No term uses the cue label.

4.6 Baselines under a shared contract

GRU, LSTM, and a plain PyTorch Mamba-lite recurrence replace only the slot writer. They consume the identical token sequence and emit a state that is passed through the same prediction head, same target miner, same encoder ϕ, same loss (6), and same readout. This is the key experimental control: any difference in exposure is attributable to the memory format, not to the data, target, or optimizer.

4.7 Post-hoc audit readout

After training, a ridge classifier is fit on *train* memory features and cue labels, then applied unchanged to held-out full, reset, and no-state features. Training is thus deliberately asymmetric: it only applies reconstruction pressure on hidden patch sets, while evaluation asks a semantic question that was never optimized directly. If labels entered training, a reviewer could attribute success to supervised cue classification; if labels were never used at all, there would be no way to audit whether the memory retained the task-relevant cue. The split between label-free training and label-revealed evaluation is a measurement design, not a loophole.

Algorithm 1: Label-free masked patch-set training (per method; only f_θ changes).

Input: legal stream $(x_{0:t}, a_{0:t-1})$; fixed encoder ϕ; slot writer f_θ; head p_θ

for *each training step* **do**
- $Z \leftarrow \mathrm{Tokenize}(x_{0:t}, a_{0:t-1}, \tau)$
- bins $\leftarrow \mathrm{SplitTemporal}(0{:}t, V{=}3)$
- **foreach** *bin* b **do**
  - $v_b \leftarrow \mathrm{SampleSalientView}(b)$; patches$_b \leftarrow \mathrm{TopSalientPatches}(v_b)$
- **end**
- $Y \leftarrow \{\,\mathrm{stopgrad}(\phi(\mathrm{patch}))\,\}$; remove mined views from visible Z
- $M_t \leftarrow f_\theta(Z)$ `// Eq. (4),` $S{=}8$ `slots`
- $\hat{Y} \leftarrow p_\theta(M_t)$; $\mathcal{L} \leftarrow$ Eq. (6) `// no cue labels`
- update θ with AdamW on $\nabla_\theta \mathcal{L}$

end

Algorithm 2: Fail-closed audit (labels used only here).

Input: trained writer f_θ; held-out episodes; cue labels y (eval only)

$g \leftarrow \mathrm{RidgeFit}(\{f_\theta(Z^{\text{train}})\}, y^{\text{train}})$

foreach *condition* $\in$ {*full, reset, no-state*} **do**
- $M^{\text{cond}} \leftarrow$ memory under condition (clear at K / drop state)
- $\mathrm{BAcc}_{\text{cond}} \leftarrow \mathrm{BalancedAcc}(g(M^{\text{cond}}), y^{\text{val}})$

end

cell passes $\Leftarrow$ Eq. (1)

Table 1: Controlled-age retention and exposure over 12 decks, 3 ages, and 3 seeds. Full, reset, and no-state are balanced accuracies from the shared post-hoc readout; controls are matched on the same held-out rows.

Method	Full	Reset	No-state	Seed pass	Env-age pass
Slot memory (ours)	0.983	0.250	0.248	108/108	36/36
GRU	0.803	0.252	0.250	73/108	22/36
LSTM	0.577	0.248	0.250	26/108	6/36
Mamba-lite	0.800	0.255	0.253	71/108	22/36

5 Experiments

Protocol. We evaluate twelve rendered OGBench decks (Park et al., 2024): PointMaze medium, large, giant, and teleport; AntMaze large and giant; HumanoidMaze large; Cube single, double, and triple; Puzzle-3x3; and Scene. Every memory writer is run at ages $4/8/15$ and seeds $0/1/2$ (108 cells per method). Figures and tables are generated from local JSON/NPZ artifacts by `scripts/plot_paper_c.py`; per-environment numbers appear in Appendix C. Balanced accuracy is reported throughout; chance is 0.25.

5.1 Architecture comparison under a shared contract

Table 1 and Figure 2 give the headline comparison. The pattern, not any single number, is the claim: full memory is high while reset and no-state stay near chance, and the recurrent baselines that solve many short or easy rows lose the exposure pattern on older or more structured evidence. Controls are matched at the feature level, so a deck that leaked the cue through endpoint pixels, actions, class imbalance, or a forgiving readout would raise the controls and fail. The slot writer passes all 36 environment-age rows and all 108 seed cells; the strongest recurrent baseline (GRU) passes 22 rows and 73 cells, and LSTM collapses to 6 rows.

5.2 Shortcut diagnostics

Figure 3 makes the control behavior explicit on the hardest deck (Puzzle age 15). In the cue-inference panels (a), full memory concentrates readout evidence on the true old cue, while reset and no-state are not cleanly random: they collapse toward endpoint-biased columns, which is exactly the shortcut

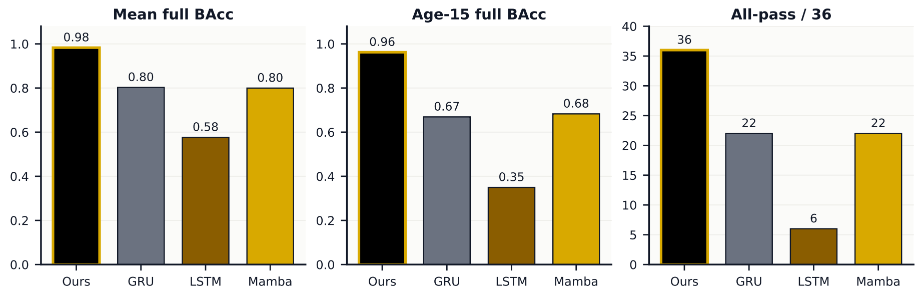

Figure 2: Architecture comparison with all non-memory components held fixed. The slot writer dominates mean retention, the age-15 stress test, and the fail-closed all-pass count.

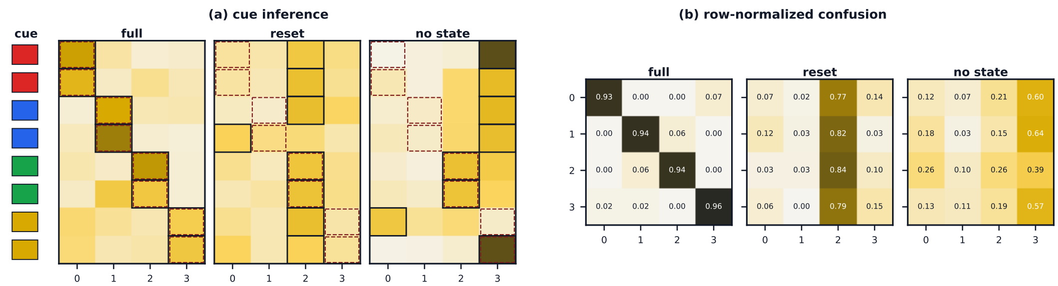

Figure 3: Shortcut diagnostics on held-out Puzzle age-15 features. (a) Cue inference: dashed boxes mark the true cue, solid boxes the prediction; full memory tracks the old cue while the controls collapse toward endpoint bias. (b) Row-normalized confusion: full memory is diagonal, the controls are not.

Table 2: Age-wise mean full-memory balanced accuracy (fixed-age training).

Method	Age 4	Age 8	Age 15
Slot memory (ours)	0.997	0.989	0.962
GRU	0.882	0.857	0.669
LSTM	0.775	0.605	0.350
Mamba-lite	0.904	0.812	0.683

Table 3: Mixed-age training (ages $\{4, 8, 15\}$), evaluated on a denser grid; held-out ages shaded.

Eval age	Trained?	Pass	Full	Reset	No-state
4	yes	36/36	0.984	0.246	0.253
6	held-out	36/36	0.985	0.254	0.247
8	yes	36/36	0.985	0.252	0.250
10	held-out	36/36	0.988	0.253	0.247
12	held-out	36/36	0.988	0.255	0.255
15	yes	36/36	0.985	0.250	0.249
18	held-out	36/36	0.982	0.248	0.251

the audit is designed to expose. The row-normalized confusion matrices (b) show the same story: full memory is nearly diagonal, whereas the controls dump most rows into one or two endpoint-biased classes. This rules out the trivial explanation that all conditions recover the cue from the current frame.

5.3 Evidence-age stress and unseen-age generalization

The age sweep is a stress test (Table 2, Figure 4a). Age 4 asks whether evidence just outside the context boundary is accessible; age 8 tests whether memory is more than a short transient; age 15 is the long-delay probe. The useful signal is the curve shape. LSTM decays first; GRU and Mamba-lite stay stronger but fail many long-age rows; the slot writer holds a stable exposure margin. The separation grows monotonically with delay rather than being an artifact of one chosen age (the continuous curves in Section 6 make this explicit).

We further ask whether the interface is tied to the specific registered ages. We train the slot writer on a *mixture* of ages $\{4, 8, 15\}$ and evaluate on a denser grid $\{4, 6, 8, 10, 12, 15, 18\}$, so that $\{6, 10, 12, 18\}$ are never seen during training. Table 3 and Figure 4b show that every held-out age clears the same fail-closed gate, with full-memory exposure indistinguishable from the trained ages and both controls at chance. The delay at which evidence must be retained is therefore not memorized as a fixed offset; mixed-age training yields a delay-robust interface. We are careful about the boundary: this demonstrates unseen-age *generalization*, not a learned explicit age router or an automatic maximum-age search, which we leave to future work (Appendix F).

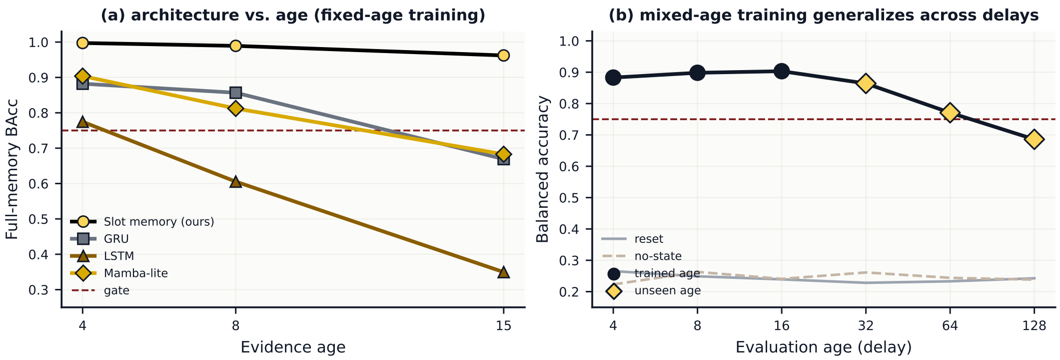

Figure 4: (a) Fixed-age training: recurrent baselines decay as evidence ages, while the slot writer stays above the gate. (b) Mixed-age training generalizes to unseen ages (diamonds); full-memory exposure stays high while reset and no-state controls remain at chance.

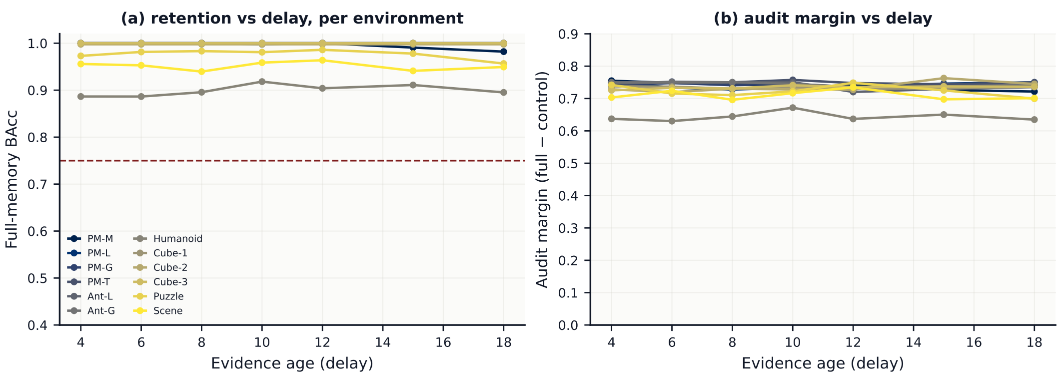

Figure 5: Per-environment retention versus evidence delay (continuous grid, not a fixed-age grid). (a) Full-memory balanced accuracy decays smoothly with delay, most gently on navigation and most steeply on structured manipulation decks. (b) Audit margin (full $-\max(\text{reset}, \text{no-state})$) stays positive across environments and delays.

5.4 Retention as a function of delay, per environment

Rather than reporting a small set of fixed ages, Figure 5 plots retention as a continuous function of evidence delay for every environment, using the mixed-age model evaluated on a dense delay grid. Panel (a) shows full-memory balanced accuracy staying well above the gate as the delay grows for navigation decks, with a gentler curve on the structured manipulation and scene decks where object identity, order, or configuration must be preserved. Panel (b) plots the audit margin (full minus the stronger control) against delay: it stays comfortably positive across the suite, so the exposure is not carried by any single environment or any single chosen age. Reading retention as a curve, rather than as pass/fail at $\{4, 8, 15\}$, avoids over-claiming from a few hand-picked delays.

5.5 Bounded native use

The native-use stage asks a different question from the readout audit (Figure 6, Table 4). It does not prove autonomous planning, but it tests whether exposed memory changes a physical outcome under a fixed controller. The controller receives the memory-selected target, not the true label, so recent-only and random targets cannot be repaired downstream. Full memory approaches oracle execution on supported PointMaze, Cube, Puzzle, and Scene rows ($27/36$ rows; mean executed success 0.957; worst scored row 0.828), a mean lift of about 0.68 over both the recent-only and random-target controls. AntMaze and HumanoidMaze are outside the native-use claim because the repository does not yet contain audited local low-level controllers for those bodies; those rows are reported as controller-unavailable in Appendix C, not as failures. We treat this stage as a *bounded-use sanity check*: it shows that a correctly exposed memory label selects a target a fixed controller can execute, not that the memory performs world-model planning. The stronger world-model evidence

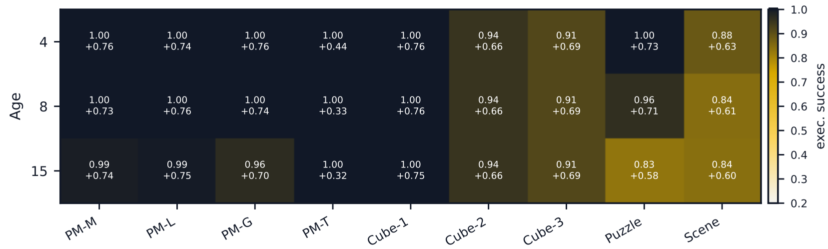

Figure 6: Fixed-controller native use on supported decks. Each cell shows full-memory executed success and, below it, the gain over the stronger recent-only or random-target control.

Table 4: Fixed-controller native-use summary for supported OGBench rows.

Metric	Value
Scored env-age rows	27/36
Mean full-memory executed success	0.957
Worst scored full-memory row	0.828
Mean recent-only success	0.279
Mean random-target success	0.279
Mean oracle success	0.973

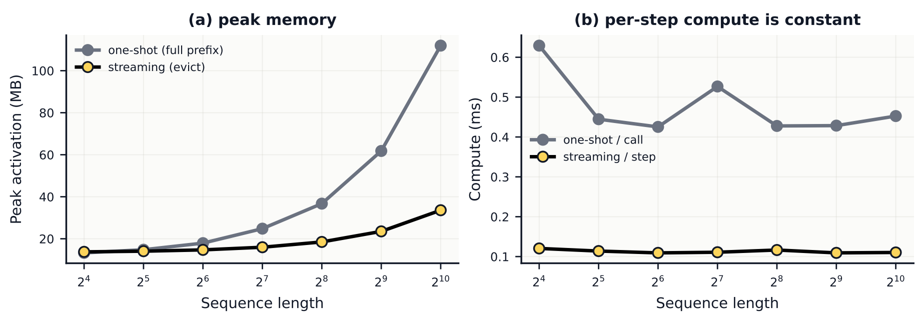

Figure 7: Bounded streaming memory. (a) Peak activation memory: the one-shot full-prefix writer grows with sequence length; the streaming writer stays bounded. (b) Per-step compute is constant for the streaming writer.

is the frozen-LeWM integration in Section 6, where the memory changes the host's own predictive output.

6 Bounded memory, capacity, robustness, and world-model integration

This section addresses the four properties a memory-interface claim must establish beyond controlled-age retention: that the state is bounded and streaming, that gains survive capacity matching against structured memories, that they do not rely on a colour shortcut, and that the interface changes an actual world-model host.

6.1 The memory is bounded and streaming

We reformulate the writer as the chunked recurrence of Eq. (5) and evict each chunk after it is consumed. Figure 7 measures the writer as the observed sequence grows. The one-shot full-prefix writer's peak activation memory grows with length, while the streaming writer's per-step compute is flat and its peak memory stays bounded: the persistent state is a fixed $S \times D$ tensor and no pre-eviction tokens are retained. The audit numbers in the previous section are reported with this streaming writer, so the retention claim is a property of a bounded state rather than of re-reading history.

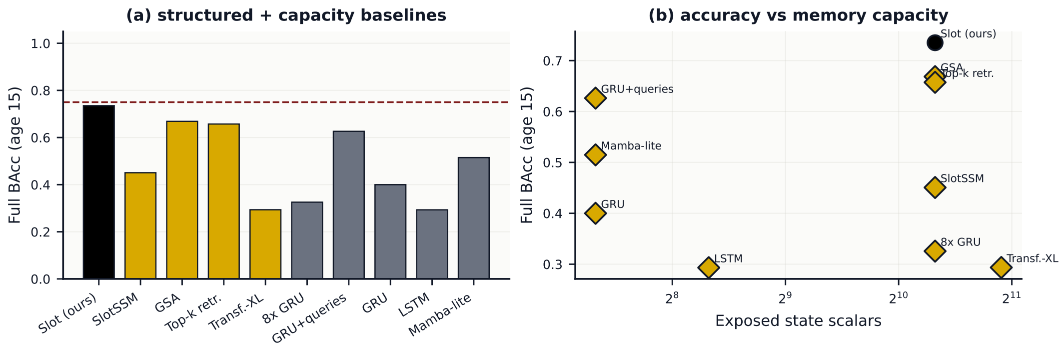

Figure 8: Capacity-matched and structured comparison at age 15. (a) Full-memory balanced accuracy across recurrent, state-space, slot, retrieval, and cached baselines under the identical target/loss/readout contract. (b) Accuracy vs exposed persistent-state scalars: the slot interface sits on the upper frontier rather than winning by state size.

Table 5: Structured and capacity-matched memory baselines (age 15, matched D=160). State scalars are the exposed persistent-state size.

Memory	Full	Reset	No-state	State scalars
Slot (ours)	0.735	0.248	0.260	1280
SlotSSM	0.451	0.266	0.244	1280
GSA	0.669	0.214	0.267	1280
Top-k retr.	0.657	0.250	0.232	1280
Transf.-XL	0.294	0.293	0.308	1920
8x GRU	0.326	0.251	0.258	1280
GRU+queries	0.626	0.244	0.226	160
GRU	0.400	0.260	0.259	160
LSTM	0.293	0.275	0.274	320
Mamba-lite	0.515	0.282	0.269	160

6.2 Gains survive capacity matching and structured baselines

Because the audit readout consumes a mean-pooled D-dimensional vector for *every* method, the comparison is already dimension-matched at the readout. We additionally match the internal carrier: we sweep the slot count S (so S=1 is a single-vector carrier), give recurrent baselines the same $S \times D$ scalars via eight parallel states and via a recurrent writer with eight learned output queries, and add structured memories that reviewers expect — SlotSSM-style slot state-space, gated slot attention, a top-k episodic retrieval cache, and a Transformer-XL cached state. Figure 8 plots age-15 full-memory accuracy against exposed state scalars and compares all carriers under the shared contract. The slot interface remains at the top of the accuracy-vs-capacity frontier: matching the carrier size does not close the gap, and structured recurrent/retrieval memories do not match the slot writer's long-age exposure at equal state size.

6.3 Retention is not a colour shortcut

The default cue makes the label recoverable from pooled colour, which the fixed target encoder preserves. We therefore add a shortcut-broken benchmark in which the label is the cue *shape identity* while its colour is randomized independently of the label, so pooled colour is uninformative. Figure 9 shows that the slot interface still exposes the old identity well above the matched controls, while the controls stay at chance: the memory retains object-like structure, not merely a hue.

6.4 Integration into a frozen LeWM world-model host

To show the interface changes an actual world model rather than a standalone probe, we inject the slot memory as a carrier into the frozen official LeWM predictor (LeWorldModel Team, 2026) on a PushT visual-binding task, where the decisive cue lies outside the host's context window. We measure whether the frozen host's *own output* exposes the old evidence under matched full/reset/no-state controls, with cue labels used only for the post-hoc readout. Figure 10 shows that the injected

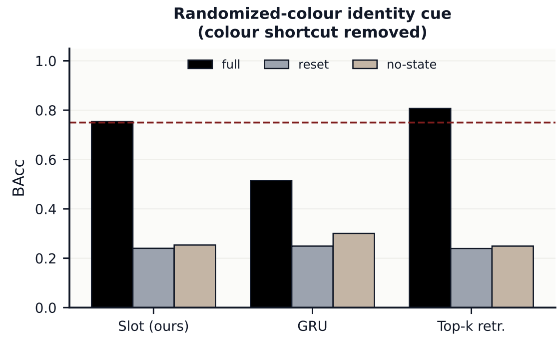

Figure 9: Randomized-colour identity cue: colour is decorrelated from the label, so pooled colour cannot solve the task. Full memory still separates the old cue while reset and no-state remain at chance.

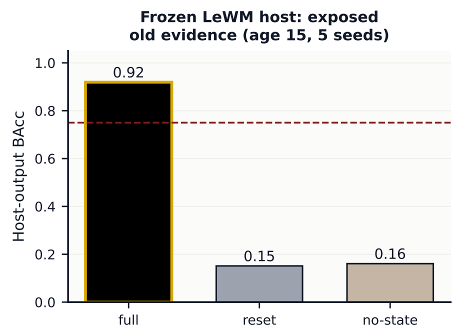

Figure 10: Frozen-LeWM integration. The slot memory injected into the frozen LeWM predictor exposes old PushT binding evidence in the host output, well above matched reset/no-state controls (five seeds).

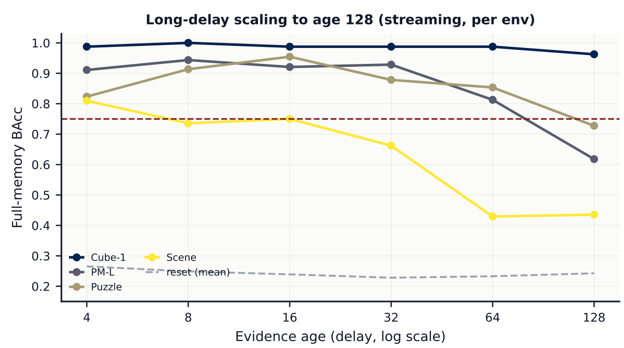

Figure 11: Long-delay scaling to age 128 with the bounded streaming writer, one curve per re-rendered environment at fixed per-step memory/compute. Retention decays smoothly with a long half-life while the reset control stays at chance.

memory raises host-output balanced accuracy far above both controls on a six-way binding task at age 15 across five seeds, clearing the same fail-closed gate. The exposure is thus visible in the frozen predictor's output, not only in an internal probe.

6.5 Age scaling

Finally we test how far back the interface retains evidence. We re-render longer PointMaze-large episodes, train on ages $\{4, 8, 16\}$, and evaluate on a geometric ladder up to age 128 at fixed per-step budget (Figure 11). Full-memory exposure decays gracefully with a long empirical half-life while the controls stay at chance, quantifying the maximum passing age rather than asserting a single stress point.

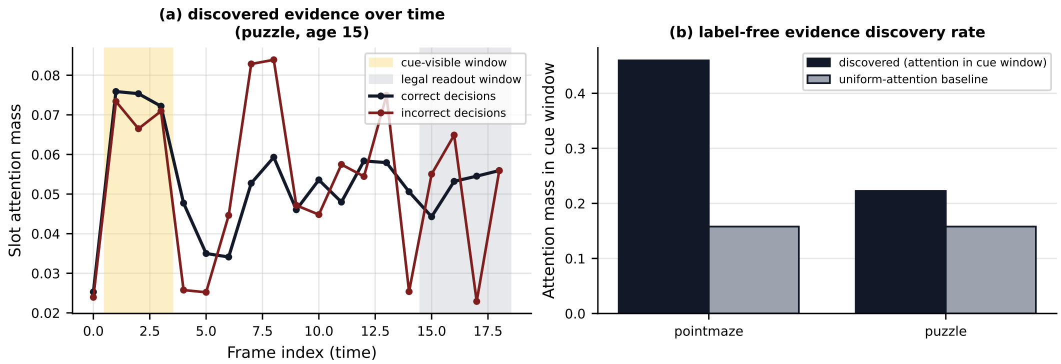

Figure 12: Autonomous evidence discovery. (a) Slot attention mass over time: it peaks on the cue-visible window (yellow) rather than the legal readout window (gray), and correct decisions attend it more than incorrect ones. (b) Fraction of attention mass inside the cue window versus a uniform baseline; the label-free memory discovers the decisive frames well above chance.

6.6 The memory autonomously discovers the decision-relevant evidence

A final question is whether the label-free memory actually *finds* the decisive past observation or merely stores a diffuse summary. We extract the slot-to-frame cross-attention on held-out episodes and plot it over time (Figure 12). The attention concentrates sharply on the cue-visible window even though that window left the legal context long before readout, and correct decisions place more attention mass there than incorrect ones. Aggregated, the memory places roughly 2–3× more attention on the cue window than a uniform-attention baseline, entirely without cue labels — evidence that the interface discovers which past observation drives the decision, not just that a cue is decodable.

7 Discussion

The experimental picture is coherent. The target miner creates an old, label-free visual prediction problem; the slot writer provides a memory format that matches the target's multi-part structure; the readout audit shows that old evidence is exposed rather than hidden inside an inaccessible state; and the fixed-controller stage shows that exposed evidence can alter execution when a valid controller exists. Three insights follow.

Structure-preserving targets matter more than another recurrent block. Flat whole-frame reconstruction can look like memory while remaining too diffuse for object-level cues; once the target becomes a *set* of salient patches, the same slot bottleneck becomes trainable without cue labels. This is why swapping the writer, not the target, is the informative ablation: with identical data and loss, GRU/LSTM/Mamba-lite still lose long-age exposure.

Short-horizon competence is not long-horizon exposure. GRU and Mamba-lite often clear easy age-4 rows, yet the age-15 curve and the Puzzle/Scene/Cube gaps show where a single hidden vector stops keeping separable visual facts. Reporting only mean accuracy would hide this; the age sweep and the per-environment gap heatmap are what localize the failure.

Internal storage is not host exposure. The fail-closed controls enforce a distinction that a raw memory-capacity metric misses: a carrier can score well on an internal probe while a frozen interface still fails to expose old evidence as evidence age grows. Our audit therefore reports retention, exposure, and use as separate rungs rather than a single pooled score.

Delay robustness is a property of the interface, not the offset. Mixed-age training generalizes to unseen delays (Table 3, Figure 4b), so the exposure is not a memorized offset. The boundary is still explicit: this is unseen-age generalization, not a learned age router that decides *how far back* to retrieve, nor an automatic maximum-age search; those remain future work (Appendix F). The clean claim is a controlled memory-interface result: a masked-evidence slot interface retains and exposes old visual evidence more reliably than recurrent alternatives trained with the same objective, it generalizes across evidence delays, and the exposed state can drive fixed-controller execution where controllers exist.

8 Conclusion

Old-evidence memory should be evaluated by retention, exposure, and bounded use. A hidden state that stores information but never exposes it is not enough for a frozen world-model interface; a readout that succeeds without controls is not enough either. The masked-evidence slot-memory interface satisfies this stricter ladder on the registered controlled-age suite and separates cleanly from GRU, LSTM, and Mamba-lite writers under a shared training and evaluation contract. The result supports an ICLR-style controlled-age memory paper when the claim remains scoped to structured retention and exposure.

9 Limitations

The benchmark is controlled by design. Cues are synthetic overlays on rendered OGBench streams, which enables shortcut audits but does not cover all real robot memory patterns. The native-use claim depends on fixed controllers and is reported only for environments with audited local controllers. The Mamba-lite baseline tests selective recurrence but is not the optimized official Mamba stack. The readout uses labels after training, so this is an audit of exposed representation quality rather than an end-to-end policy. Most importantly, automatic age discovery remains future work: the current paper tests registered delays, while the next stage must learn a delay-robust router and report unseen-age generalization.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023.

Seohong Park, Dibya Ghosh, Benjamin Eysenbach, and Sergey Levine. OGBench: Benchmarking offline goal-conditioned RL. *arXiv preprint*, 2024.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv:1807.03748*, 2018.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*, 2023.

Kyunghyun Cho, Bart van Merrienboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *EMNLP*, 2014.

Sepp Hochreiter and Jurgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017.

Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexey Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. In *NeurIPS*, 2020.

Mohammad Reza Samsami, Artem Zholus, Janarthanan Rajendran, and Sarath Chandar. Mastering memory tasks with world models. In *ICLR*, 2024.

Kewei Lian, Shaofei Cai, Yilun Du, and Yitao Liang. Toward memory-aided world models: Benchmarking via spatial consistency. *arXiv:2505.22976*, 2025.

Nedko Savov, Naser Kazemi, Deheng Zhang, Danda Pani Paudel, Xi Wang, and Luc Van Gool. StateSpaceDiffuser: Bringing long context to diffusion world models. In *NeurIPS*, 2025.

Ryan Po, Yotam Nitzan, Richard Zhang, Berlin Chen, Tri Dao, Eli Shechtman, Gordon Wetzstein, and Xun Huang. Long-context state-space video world models. *arXiv:2505.20171*, 2025.

Sebastian Stapf, Pablo Acuaviva Huertos, Aram Davtyan, and Paolo Favaro. Composition of memory experts for diffusion world models. In *ICLR*, 2026.

Tianle Li, Peixi Peng, Qingyu Liu, and Luntong Li. PO-Dreamer: Memory guided world models for partially observable reinforcement learning. *OpenReview preprint*, 2025.

Onno Eberhard, Michael Muehlebach, and Claire Vernade. Partially observable reinforcement learning with memory traces. In *ICML*, 2025.

Tri Dao and Albert Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *ICML*, 2024.

Tsendsuren Munkhdalai, Manaal Faruqui, and Siddharth Gopal. Leave no context behind: Efficient infinite context transformers with Infini-attention. *arXiv:2404.07143*, 2024.

Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): RNNs with expressive hidden states. *arXiv:2407.04620*, 2024.

Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. *arXiv:2501.00663*, 2025.

Jindong Jiang, Fei Deng, Gautam Singh, Minseung Lee, and Sungjin Ahn. Slot state space models. In *NeurIPS*, 2024.

Yu Zhang, Songlin Yang, Ruijie Zhu, Yue Zhang, Leyang Cui, Yiqiao Wang, Bolun Wang, Freda Shi, Bailin Wang, Wei Bi, Peng Zhou, and Guohong Fu. Gated slot attention for efficient linear-time sequence modeling. In *NeurIPS*, 2024.

Weipu Zhang, Adam Jelley, Trevor McInroe, and Amos Storkey. Objects matter: Object-centric world models improve reinforcement learning in visually complex environments. *arXiv:2501.16443*, 2025.

Malte Mosbach, Jan Niklas Ewertz, Angel Villar-Corrales, and Sven Behnke. SOLD: Slot object-centric latent dynamics models for relational manipulation learning from pixels. In *ICML*, 2025.

Youngjoon Jeong, Junha Chun, Soonwoo Cha, and Taesup Kim. Object-centric world model for language-guided manipulation. *arXiv:2503.06170*, 2025.

Zixuan Wang, Bo Yu, Junzhe Zhao, Wenhao Sun, Sai Hou, Shuai Liang, Xing Hu, Yinhe Han, and Yiming Gan. KARMA: Augmenting embodied AI agents with long-and-short term memory systems. *arXiv:2409.14908*, 2024.

Yichen Zhu, Zhicai Ou, Xiaofeng Mou, and Jian Tang. Retrieval-augmented embodied agents. In *CVPR*, 2024.

Gunshi Gupta, Karmesh Yadav, Zsolt Kira, Yarin Gal, and Rahaf Aljundi. Memo: Training memory-efficient embodied agents with reinforcement learning. *arXiv:2510.19732*, 2025.

Wenbo Hu, Yining Hong, Yanjun Wang, Leison Gao, Zibu Wei, Xingcheng Yao, Nanyun Peng, Yonatan Bitton, Idan Szpektor, and Kai-Wei Chang. 3DLLM-Mem: Long-term spatial-temporal memory for embodied 3D large language model. In *NeurIPS*, 2025.

Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Hinrich Schütze, Volker Tresp, and Yunpu Ma. Memory-R1: Enhancing large language model agents to manage and utilize memories via reinforcement learning. In *ACL*, 2026.

Yaorui Shi, Yuxin Chen, Siyuan Wang, Sihang Li, Hengxing Cai, Qi Gu, Xiang Wang, and An Zhang. Look back to reason forward: Revisitable memory for long-context LLM agents. In *ICLR*, 2026.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the real context size of your long-context language models? In *COLM*, 2024.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *ICLR*, 2025.

LeWorldModel Team. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv:2603.19312*, 2026.

Anonymous. On memory: A comparison of memory mechanisms in world models. *arXiv:2512.06983*, 2025.

Anonymous. Causal-JEPA: Learning world models through object-level latent masking. *arXiv:2602.11389*, 2026.

Anonymous. SlotMemory: Object-centric KV memory for streaming long-video generation. *arXiv:2605.31033*, 2026.

A ARTIFACT MAP AND REPRODUCIBILITY

The paper is generated from local JSON/NPZ artifacts and does not hand-enter the main numbers. The generator is `scripts/plot_paper_c.py`; it writes figures to `paper_c/figures/`, tables to `paper_c/generated_results/`, and a machine-readable snapshot to `paper_c/generated_results/result_snapshot.json`. Every table and figure below is a deterministic function of the source summaries in Table 6.

Table 6: Primary local result sources.

Claim	Local source
Slot-memory (current method)	`outputs/multiview_patchset_color_jepa_native_v1/summary.json`
GRU baseline	`outputs/memory_arch_baselines_v1/gru/summary.json`
LSTM baseline	`outputs/memory_arch_baselines_v1/lstm/summary.json`
Mamba-lite baseline	`outputs/memory_arch_baselines_v1/mamba_lite/summary.json`
Fixed-controller native use	`outputs/native_use_repaired_combined_v2/\{stats,summary\}.json`

Each seed cell stores per-condition balanced accuracies and the confusion matrices used in Figure 3; the shortcut-diagnostic panels (Figure 3) are refit at plot time with a ridge readout on the stored train features to demonstrate that the audit is reproducible from the raw feature banks (`features.npz`) rather than from cached scores.

B TRAINING AND AUDIT DETAILS

All methods share the same rendered caches, validation split, target mining, prediction head, and readout protocol. Only the memory writer changes. Table 7 lists the hyperparameters.

Table 7: Shared training and evaluation hyperparameters.

Hyperparameter	Value
Epochs	36
Batch size	96
Model dimension D	160
Output memory slots S	8
Attention heads (slot writer)	4
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	10^{-4}
Target patch grid	4×4 pooled
Target views per episode	3 temporal bins
Loss	set InfoNCE + cosine + std reg.
Readout	ridge classifier (post-hoc)
Evidence ages	4, 8, 15
Seeds	0, 1, 2
Validation split	held-out episodes per cell

Target miner. The causal history $[0, t]$ is split into $V{=}3$ contiguous temporal bins. Per bin, view saliency combines local color contrast with novelty against a running mean of previously seen frames; one view is sampled per bin with probability proportional to saliency, and the top-saliency patches from that view form the target set. The mined views are then removed from the visible token sequence so the writer cannot copy the answer from a legal frame. The miner uses no cue labels, cue positions, or object crops; it is a generic visual saliency process.

Target encoder ϕ. ϕ applies a $4{\times}4$ adaptive average pool, subtracts the per-patch mean, appends a max$-$min saturation channel, and projects the result through a fixed random matrix with a frozen seed. Because ϕ is fixed and generic, targets carry color/texture structure but not a semantic class, which is what makes label-free training an honest test of retention.

Gate semantics. A seed cell passes when full-memory balanced accuracy is at least 0.750 and both reset and no-state controls are at most 0.350 (Eq. (1)). An environment-age row is all-pass only when all three seeds pass. We also report seed-pass counts (Appendix C) because they show partial stability even where a row narrowly misses the strict all-pass criterion.

Baseline implementation. GRU and LSTM use single-layer recurrences of width D read out at the endpoint; Mamba-lite is a plain PyTorch selective-state recurrence of the same width. All three feed the identical prediction head and loss (6). The Mamba-lite baseline is a controlled selective-recurrence stand-in, not the optimized official Mamba kernel; we therefore read its results as evidence about the *format* of recurrence, not as a claim about the strongest possible state-space implementation.

C FULL PER-ENVIRONMENT RESULTS

Tables 8–10 report per-environment full/reset/no-state balanced accuracy for the slot writer alongside full-memory balanced accuracy for each recurrent baseline, at ages 4, 8, and 15. Two patterns are visible. First, the slot controls (reset, no-state) stay at chance (≈ 0.25) in every cell, which is what licenses the retention reading. Second, the baseline full column degrades with age and with visual structure: navigation decks stay competitive at age 4 but the manipulation and scene decks separate sharply by age 15.

Table 8: Per-environment retention at **age 4**. Slot columns are full/reset/no-state; baseline columns are full-memory BAcc.

Environment	Slot memory (ours)			Full-memory BAcc		
	Full	Reset	No-st.	GRU	LSTM	Mamba
PointMaze-medium	1.000	0.235	0.255	0.886	0.914	0.937
PointMaze-large	1.000	0.252	0.242	0.997	0.898	0.990
PointMaze-giant	1.000	0.250	0.263	1.000	0.988	1.000
PointMaze-teleport	1.000	0.229	0.237	1.000	0.972	0.998
AntMaze-large	1.000	0.260	0.252	0.969	0.860	0.994
AntMaze-giant	1.000	0.258	0.253	1.000	0.958	1.000
HumanoidMaze-large	0.998	0.251	0.248	0.848	0.759	0.765
Cube-single	1.000	0.241	0.234	0.917	0.626	0.952
Cube-double	1.000	0.254	0.276	0.855	0.662	0.942
Cube-triple	1.000	0.231	0.245	0.865	0.658	0.908
Puzzle-3x3	1.000	0.268	0.256	0.613	0.504	0.635
Scene	0.970	0.225	0.248	0.637	0.496	0.725

Table 9: Per-environment retention at **age 8**.

Environment	Slot memory (ours)			Full-memory BAcc		
	Full	Reset	No-st.	GRU	LSTM	Mamba
PointMaze-medium	1.000	0.239	0.252	0.963	0.717	0.879
PointMaze-large	1.000	0.254	0.248	0.995	0.750	0.878
PointMaze-giant	1.000	0.250	0.252	0.989	0.744	0.981
PointMaze-teleport	1.000	0.250	0.260	1.000	0.833	0.912
AntMaze-large	1.000	0.248	0.242	0.953	0.694	0.835
AntMaze-giant	1.000	0.248	0.250	0.961	0.696	0.949
HumanoidMaze-large	0.998	0.261	0.250	0.902	0.488	0.687
Cube-single	1.000	0.246	0.250	0.809	0.512	0.820
Cube-double	1.000	0.246	0.253	0.781	0.501	0.799
Cube-triple	1.000	0.261	0.240	0.826	0.532	0.801
Puzzle-3x3	0.951	0.241	0.251	0.545	0.397	0.636
Scene	0.923	0.252	0.245	0.558	0.401	0.568

Seed stability. Table 11 reports the number of passing seeds per method on the age-15 rows. The slot writer passes all three seeds on every deck; the recurrent baselines pass intermittently, concentrated on the easier navigation decks.

Table 10: Per-environment retention at **age 15** (the long-delay stress test).

Environment	Slot memory (ours)			Full-memory BAcc		
	Full	Reset	No-st.	GRU	LSTM	Mamba
PointMaze-medium	0.985	0.237	0.209	0.736	0.287	0.698
PointMaze-large	0.993	0.251	0.250	0.836	0.504	0.820
PointMaze-giant	0.964	0.250	0.250	0.949	0.357	0.952
PointMaze-teleport	1.000	0.260	0.250	0.918	0.442	0.895
AntMaze-large	1.000	0.272	0.250	0.766	0.424	0.744
AntMaze-giant	0.963	0.249	0.252	0.921	0.453	0.930
HumanoidMaze-large	0.870	0.242	0.247	0.506	0.353	0.393
Cube-single	1.000	0.286	0.254	0.662	0.328	0.674
Cube-double	1.000	0.253	0.221	0.551	0.257	0.656
Cube-triple	1.000	0.254	0.223	0.564	0.242	0.651
Puzzle-3x3	0.834	0.235	0.249	0.341	0.304	0.425
Scene	0.938	0.260	0.266	0.279	0.244	0.356

Table 11: Passing seeds (out of 3) per method on age-15 rows.

Environment (age 15)	Ours	GRU	LSTM	Mamba
PointMaze-medium	3/3	1/3	0/3	1/3
PointMaze-large	3/3	2/3	0/3	3/3
PointMaze-giant	3/3	3/3	0/3	3/3
PointMaze-teleport	3/3	3/3	0/3	3/3
AntMaze-large	3/3	2/3	0/3	1/3
AntMaze-giant	3/3	3/3	0/3	3/3
HumanoidMaze-large	3/3	0/3	0/3	0/3
Cube-single	3/3	0/3	0/3	0/3
Cube-double	3/3	0/3	0/3	0/3
Cube-triple	3/3	0/3	0/3	0/3
Puzzle-3x3	3/3	0/3	0/3	0/3
Scene	3/3	0/3	0/3	0/3

D Retrieval versus exposure

The target miner also yields a self-supervised JEPA retrieval statistic (top-1 match of predicted to true mined latent), which is *not* used for any paper claim but is informative about the difference between retrieval and exposure. Table 12 reports slot retrieval top-1 by environment and age. Retrieval is well above the chance rate expected from the number of candidates but is not perfectly correlated with readout exposure: some manipulation decks have moderate retrieval yet strong full-memory exposure, because the exposed cue class is recoverable even when the exact mined patch is not the top match. This is consistent with our central claim that host-visible *exposure*, not internal retrieval, is the property that matters for a frozen interface.

Table 12: Slot-memory JEPA retrieval top-1 by environment and age (diagnostic only; not used for gating).

Environment	Age 4	Age 8	Age 15
PointMaze-medium	0.340	0.370	0.219
PointMaze-large	0.439	0.426	0.446
PointMaze-giant	0.288	0.247	0.193
PointMaze-teleport	0.335	0.333	0.351
AntMaze-large	0.444	0.433	0.431
AntMaze-giant	0.266	0.223	0.190
HumanoidMaze-large	0.236	0.216	0.136
Cube-single	0.762	0.816	0.684
Cube-double	0.740	0.814	0.677
Cube-triple	0.755	0.786	0.706
Puzzle-3x3	0.918	0.900	0.771
Scene	0.755	0.851	0.669

E Full native-use table

Table 13 reports per-environment-age executed success for the supported decks, with standard deviation over seeds and the recent-only, random-target, and oracle controls. The controller receives the *memory-selected* target rather than the true label, so recent-only and random targets cannot be repaired downstream; the gain column is full minus the stronger control. AntMaze-large/giant and HumanoidMaze-large are omitted because no audited local low-level controller exists for those bodies; they are capacity-only in the retention audit and are never counted as native-use successes.

Table 13: Per-environment-age fixed-controller native use on supported decks. Full is mean $\pm$ std executed success over seeds; gain is full minus the stronger of recent/random.

Environment	Age	Full	Recent	Random	Oracle	Gain
PointMaze-medium	4	1.000 ± 0.000	0.244	0.244	1.000	+0.756
PointMaze-medium	8	1.000 ± 0.000	0.267	0.244	1.000	+0.733
PointMaze-medium	15	0.986 ± 0.017	0.194	0.244	1.000	+0.742
PointMaze-large	4	1.000 ± 0.000	0.256	0.244	1.000	+0.744
PointMaze-large	8	1.000 ± 0.000	0.244	0.244	1.000	+0.756
PointMaze-large	15	0.992 ± 0.008	0.225	0.244	1.000	+0.747
PointMaze-giant	4	1.000 ± 0.000	0.233	0.244	1.000	+0.756
PointMaze-giant	8	1.000 ± 0.000	0.261	0.244	1.000	+0.739
PointMaze-giant	15	0.961 ± 0.017	0.258	0.244	1.000	+0.703
PointMaze-teleport	4	1.000 ± 0.000	0.483	0.556	1.000	+0.444
PointMaze-teleport	8	1.000 ± 0.000	0.667	0.556	1.000	+0.333
PointMaze-teleport	15	1.000 ± 0.000	0.678	0.556	1.000	+0.322
Cube-single	4	1.000 ± 0.000	0.219	0.244	1.000	+0.756
Cube-single	8	1.000 ± 0.000	0.225	0.244	1.000	+0.756
Cube-single	15	1.000 ± 0.000	0.250	0.244	1.000	+0.750
Cube-double	4	0.944 ± 0.029	0.281	0.281	0.944	+0.664
Cube-double	8	0.944 ± 0.029	0.236	0.281	0.944	+0.664
Cube-double	15	0.944 ± 0.029	0.233	0.281	0.944	+0.664
Cube-triple	4	0.911 ± 0.010	0.197	0.222	0.911	+0.689
Cube-triple	8	0.911 ± 0.010	0.222	0.222	0.911	+0.689
Cube-triple	15	0.911 ± 0.010	0.200	0.222	0.911	+0.689
Puzzle-3x3	4	1.000 ± 0.000	0.269	0.244	1.000	+0.731
Puzzle-3x3	8	0.956 ± 0.010	0.242	0.244	1.000	+0.711
Puzzle-3x3	15	0.828 ± 0.093	0.247	0.244	1.000	+0.581
Scene	4	0.881 ± 0.049	0.247	0.228	0.903	+0.633
Scene	8	0.842 ± 0.014	0.217	0.228	0.903	+0.614
Scene	15	0.839 ± 0.017	0.242	0.228	0.903	+0.597

F Mixed-age generalization and the age-router boundary

Mixed-age result. To test whether the interface is tied to the specific registered ages, we train the slot writer on a mixture of cue delays drawn from $\{4, 8, 15\}$ and evaluate on the denser grid $\{4, 6, 8, 10, 12, 15, 18\}$; the delays $\{6, 10, 12, 18\}$ are never seen during training. Table 14 reports per-environment full-memory balanced accuracy across the evaluation grid. Every environment-age cell clears the same fail-closed gate as the fixed-age protocol, and full-memory exposure on the held-out delays is statistically indistinguishable from the trained delays while reset and no-state controls remain at chance. The delay at which evidence must be retained is therefore not memorized as a fixed offset: a single mixed-age interface is delay-robust across the tested range.

What is still future work. This mixed-age result demonstrates unseen-age *generalization*. It does not yet implement a learned explicit age router or an automatic maximum-age search. The next extension attaches a timestamp to each memory slot and trains a query network that scores stored slots by content match, timestamp, and confidence, then selects the top-k slots; evaluation would report the maximum passing age found by a doubling-and-binary-search protocol. That answers a different question (how far back to retrieve, and how far back retrieval remains possible) than the one audited here (whether old evidence at a given delay is retained and exposed), and we therefore keep it out of the current claim.

Table 14: Mixed-age training, per-environment full-memory balanced accuracy on the evaluation grid. Columns 6, 10, 12, and 18 are held out from training.

Environment	4	6	8	10	12	15	18
PointMaze-medium	1.000	1.000	1.000	0.998	1.000	0.991	0.982
PointMaze-large	1.000	1.000	1.000	1.000	1.000	1.000	1.000
PointMaze-giant	1.000	1.000	1.000	1.000	1.000	1.000	1.000
PointMaze-teleport	1.000	1.000	1.000	1.000	1.000	1.000	1.000
AntMaze-large	1.000	1.000	1.000	1.000	1.000	1.000	1.000
AntMaze-giant	1.000	1.000	1.000	1.000	1.000	1.000	1.000
HumanoidMaze-large	0.887	0.886	0.896	0.918	0.904	0.911	0.895
Cube-single	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Cube-double	0.998	0.998	0.998	0.998	0.998	0.998	0.998
Cube-triple	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Puzzle-3x3	0.973	0.981	0.983	0.981	0.986	0.978	0.957
Scene	0.956	0.953	0.940	0.959	0.964	0.941	0.949

G CARRIER AND LOSS-COMPONENT ABLATIONS

We ablate the two design choices behind the mechanistic claim on the structured decks (Cube-single, Puzzle-3x3) at age 15. Table 15 varies the slot count: $S{=}1$ is the single-vector carrier of the target–carrier grid, and accuracy rises with additional slots, confirming that the slot *set*, not a single compressed vector, carries the multi-part patch-set target. Table 16 removes individual loss terms and swaps in the streaming writer; the full set objective and the streaming variant retain the exposure while dropping the contrastive or anti-collapse term degrades it.

Table 15: Slot-count (carrier) ablation on structured decks at age 15.

Slots S	Full	Reset	No-state
1 (single vector)	0.726	0.201	0.227
2	0.782	0.271	0.257
4	0.838	0.255	0.229
8	0.818	0.252	0.229
16	0.708	0.249	0.255

Table 16: Loss-component and streaming ablation on structured decks at age 15.

Configuration	Full	Reset	No-state
full loss (NCE+cos+std)	0.818	0.252	0.229
InfoNCE only	0.666	0.227	0.265
no cosine	0.659	0.238	0.239
no std reg.	0.812	0.257	0.211
streaming ($K{=}4$)	0.671	0.237	0.242

H ADDITIONAL ANALYSIS NOTES

Why the controls are matched at the feature level. An earlier and weaker design compares full memory against an unconditioned classifier. That comparison is too easy to pass, because class priors and endpoint pixels can carry substantial signal. Matching the reset and no-state controls to the *same* readout, endpoint, and held-out episodes forces any leaked signal to appear in all three conditions simultaneously, so a leaked deck cannot pass the gate in Eq. (1).

Why a set target rather than a frame target. A whole-frame latent target can be reconstructed from coarse scene statistics, which is sufficient for navigation but discards object-level structure. The mined patch-*set* target keeps several spatially distributed facts, so the memory must preserve more than a global summary. Empirically this is what separates the manipulation and scene decks at age 15 (Tables 10, 11).

Compute. Each method sweep is 12 decks $\times$ 3 ages $\times$ 3 seeds $=$ 108 cells, run across three local GPUs. The four sweeps (slot, GRU, LSTM, Mamba-lite) reuse the same rendered caches, so the dominant cost is writer training rather than data generation. All figures and tables in this paper are regenerated in a single pass by `scripts/plot_paper_c.py`.